

# RS003 Extension — Design Your Game Art with AI

**Topic:** Prompting Google Gemini for 32-bit Game Sprites · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This is a bonus mission. Your job this week is **not coding** — it is being the **art director**. So far your game uses shapes you drew with code. Now you write careful instructions (a **prompt**) and Google Gemini draws **32-bit style** sprites for your game — 3 enemy ships, 1 player ship, and 1 background. A good prompt is like a good recipe: the clearer you are, the better the picture.

 Words to know: **prompt, sprite, 32-bit, transparent background, asset**

 Open Gemini together with a grown-up or your teacher. Always check the picture before you use it.

## 1 · What Makes a Good Prompt?

A computer artist cannot read your mind. It only knows what you **say**. A strong sprite prompt names **six things**:

| Part              | Question it answers | Example                                  |
|-------------------|---------------------|------------------------------------------|
| <b>Subject</b>    | What is it?         | a tiny alien fighter ship                |
| <b>Style</b>      | What art look?      | 32-bit pixel art, retro arcade           |
| <b>View</b>       | From what angle?    | top-down view, seen from above           |
| <b>Colours</b>    | What palette?       | purple and green, glowing                |
| <b>Background</b> | What is behind it?  | plain transparent background, no scenery |
| <b>Use</b>        | Where will it go?   | a game sprite for a 2D shooter           |

**Read this prompt. Underline the six parts above inside it:**

*"A tiny alien fighter ship, 32-bit pixel art, retro arcade style, top-down view seen from above, purple and green with glowing lights, on a plain transparent background, as a game sprite for a 2D shooter."*

Why do we ask for a *transparent background*? (Hint: think about putting the ship on top of your own sky.)

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## 2 · The Prompt Recipe 🍷

Here is your fill-in-the-blank recipe. Copy it, swap the blanks for your own ideas, and you have a prompt. The words in **bold** stay the same every time — they are what make it look like a game sprite.

```
A _____ (what it is),  
in 32-bit pixel art, retro arcade style,  
top-down view seen from above,  
_____ colours (your palette),  
on a plain transparent background, no scenery,  
as a single game sprite for a 2D Pygame shooter.
```

**Fill the recipe for ONE idea right now — your player ship:**

```
A _____  
in 32-bit pixel art, retro arcade style, top-down view seen from above,  
_____ colours,  
on a plain transparent background, no scenery, as a single game sprite.
```

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### 3 · Spot the Weak Prompt 🐉

Each prompt below would give a messy result. Say what is missing and rewrite it using the recipe.

**Prompt A** — *"a spaceship"*

**What is missing?**

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**Your better prompt:**

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**Prompt B** — *"a cool enemy ship with an amazing detailed realistic 3D background and explosions and a planet and stars everywhere"*

**Hint:** you only want the ship by itself. A busy background is hard to drop into your game.

**Your better prompt:**

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**Prompt C** — *"make 5 ships"*

**Hint:** ask for **one** sprite at a time so each picture is clean and the right size.

**Your better prompt:**

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## 4 · Design Your Five Assets

Plan all five **before** you open Gemini. Give each enemy its own personality through colour and shape words (spiky, round, heavy, fast).

### Player ship

A \_\_\_\_, in 32-bit pixel art, top-down view, \_\_\_\_ colours, transparent background, single game sprite.

### Enemy ship 1

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### Enemy ship 2

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### Enemy ship 3

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### Background

The background is the one asset that should **fill the whole picture** (no transparent part). Name a mood and a few details.

**Hint:** "a deep space night sky background, 32-bit pixel art, dark blue with tiny stars and a far away planet, wide 2D game background, top-down view, no ships."

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## 5 · Make It 📷

 **Generate and save all five assets**

1. Open Google Gemini with your teacher or a grown-up.
2. Paste one prompt at a time. If the picture is wrong, **change one word** and try again — that is real prompt engineering.
3. Save each finished picture (for example `ship.png` , `enemy1.png` , `enemy2.png` , `enemy3.png` , `background.png` ).
4. You should end with **5 images**: 1 player ship, 3 enemy ships, and 1 background.

💡 No coding this week. Your job is the art only — in Week 16 you swap these pictures into the game with `pygame.image.load(...)`.

Send all **5 saved pictures**. Then explain what you did — write 4 to 6 sentences. Use these starters:

The prompt I was most proud of was ...

I had to change a word when ...

I asked for a transparent background because ...

The enemy ship I made look the scariest is the one that ...

One thing the AI got wrong was ...

If I had more time, I would ...

## 6 · Record Your Walkthrough 🎥

Record a short video of **yourself** talking through your art. You are not showing a game — you are explaining your **prompts** and **why** you chose the words you did. Try to use these words: **prompt, sprite, 32-bit, transparent background, asset**.

1. Show your 5 pictures one at a time.
2. Read one prompt out loud and point to each of the six parts (subject, style, view, colours, background, use).
3. Explain **why** you chose those colours and words for that ship.
4. Show one picture the AI got wrong and the word you changed to fix it.

**Write what you will say in your video.** Plan it here before you record.

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**Submit** 

Send your **5 pictures** + your walkthrough video to teacher on **KakaoTalk**.